Connecticut Mirror Data Dashboard

Nicole Choi Zoe Khan Bayansulu Tulepbayeva
Ellie Zdancewic

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The American Community Survey (ACS) is the most detailed public source of demographic and economic data for Connecticut, yet reporters rarely use it because the Census Bureau's tools are hard to navigate, organize data by county instead of by town, and offer little historical context. We build an interactive tool, in partnership with Connecticut (CT) Mirror, that makes ACS accessible to Connecticut reporters. We restrict the ACS to Connecticut and its 169 towns, organize roughly 1,200 concepts into 14 reporting beats, flag high-priority variables, and display historical trends alongside current estimates. Our tool offers a model for adapting large federal datasets to the needs of a specific newsroom.

1 Introduction

The Connecticut (CT) Mirror is a Pulitzer-prize winning non-profit newspaper based in Hartford, Connecticut(Pazniokas 2026). Through careful reporting of issues spanning immigration, economics, criminal justice, and public health, the CT Mirror fills a crucial gap in state policy reporting in Connecticut. Beat reporters frequently integrate Connecticut census data in their reporting to provide context on housing trends, education outcomes, and demographic statistics. However, CT Mirror reporters lack the technical background to make full use of this rich dataset. According to CT Mirror reporter Angela Eichhorst, reporters "want to be able to look at historical trends and also compare town by town but they don't know how to use the US 5 year census data estimates." Furthermore some "reporters don't necessarily know all of the data that is available to them to look at" (Angela Eichhorst 2026). Therefore, the goals of the CT Mirror Census Data Dashboard are twofold: first, to provide a user-friendly simplified tool to engage with historical and town-by-town census data, and second, to introduce reporters to the full scope of census data available.

1.1 Relevant work

Due to the high-visibility and government sponsorship of the U.S. Census, there already exist several modes of access to Census data. The Census Bureau publishes their American Consumer Survey microdata as interactive and filterable spreadsheets, accessible on their website. However, effective use of these spreadsheets requires significant background knowledge on Census variables, and it is not accessible to the average user. Additionally, while the raw data is accessible from the interactive spreadsheets, there are no visualizations.

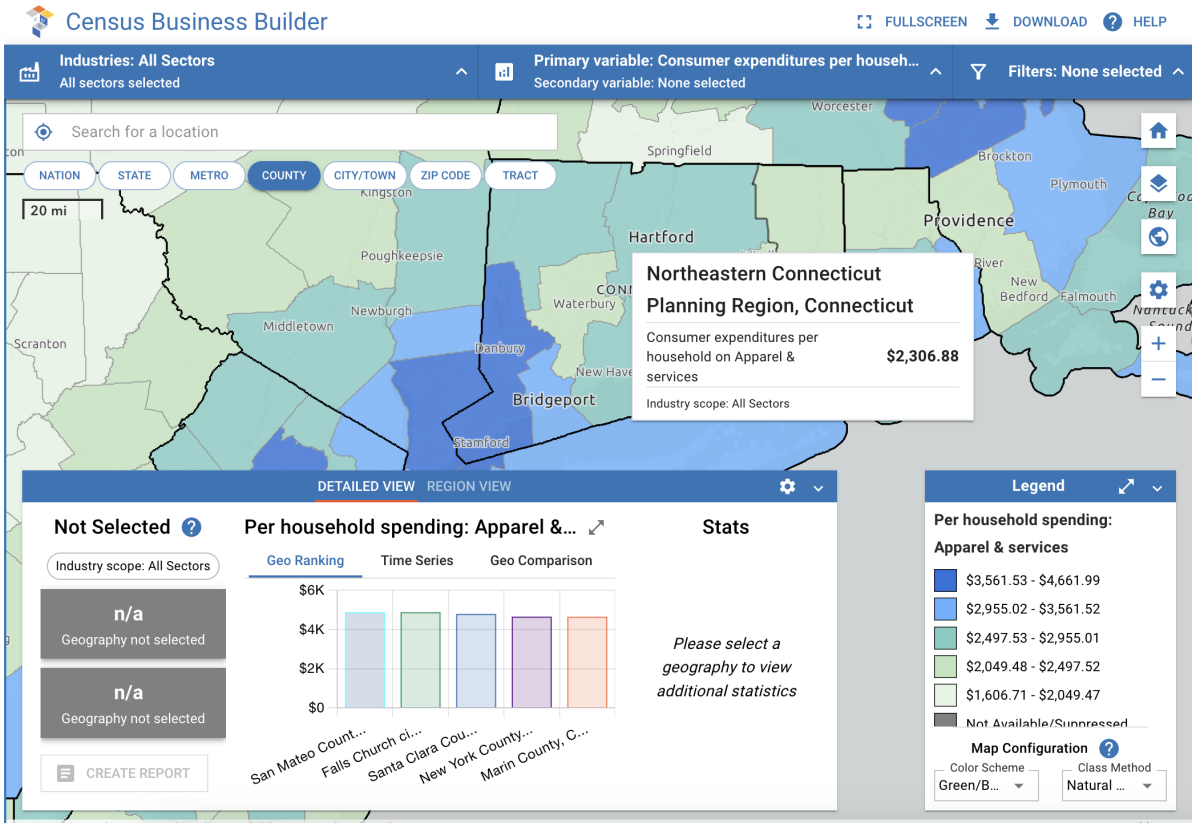


Figure 1: Census Business Builder User Interface

Alternatively, the Census Bureau publishes a data analytics tool called the Census Business Builder(US Census Bureau 2026c). This dashboard, geared towards small business owners, visualizes census data across a variety of geographies, breaking it down by a primary and secondary variable as well as a demographic filter. Figure 1 shows a screenshot of the Census Business Builder User-Interface with one primary variable selected at the county level. But while the Census Business Builder is comprehensive, it does not fit the specific needs of the CTMirror newsroom. The

dashboard has a steep learning curve and is not intuitive for a non-technical user. Additionally, it is not built for a Connecticut-specific use case. The Census Business Builder has data on the state, county, and town level. Connecticut, however, doesn't make legislative decisions on the county-level, making State and Town the only necessary geographies for this region (Angela Eichhorst 2026). Finally, the Census Business Builder does not provide historical time-series data.

1.2 Our contribution

This paper presents a new user-friendly, interactive Census Data tool that is specifically tailored to the needs of reporters in Connecticut. Unlike existing Census data exploration tools, the Connecticut Mirror Data Dashboard delivers a time-series analysis of variable trends over time. Additionally, the final dashboard is streamlined to provide reporters with the most relevant information as quickly as possible, and leaving out details that would contribute to user confusion.

2 Data

The U.S. Census Bureau authors the American Community Survey (ACS), a publicly available data set used widely by the government (and other stakeholders) to understand citizens and make data-driven decisions. It comprises an ongoing survey started in 2005 to collect information about demographics, education, economics, housing, and more, from sampled households across the country (US Census Bureau 2026a). About 3.5 million residential addresses are sampled yearly (US Census Bureau 2026b). There are different types of ACS releases, including 1-Year Estimates, which contains 12-month data for areas with populations of 65,000+ individuals, and 5-Year Estimates, which merges 60-month data for areas of all population sizes (US Census Bureau 2026b).

All ACS data covers four main categories: Social Characteristics, Housing Characteristics, Economic Characteristics, Demographic and Housing Estimates (US Census Bureau 2026a). The ACS organizes its data as a tree. Concepts form branches and name a broad table topic, such as Median Household Income in the Past 12 Months. Each concept divides into subconcepts that break the topic down by categories, such as age group, race, or sex. The leaves are individual variables, each of which holds one estimate. The full ACS contains approximately 1,200 concepts and 28,500 subconcepts.

For this project, we restrict our dataset to Connecticut. We keep only records for the state and its 169 towns. The geographic location and 2024 total population in these towns is displayed in Figure 2. This restriction removes the national scope

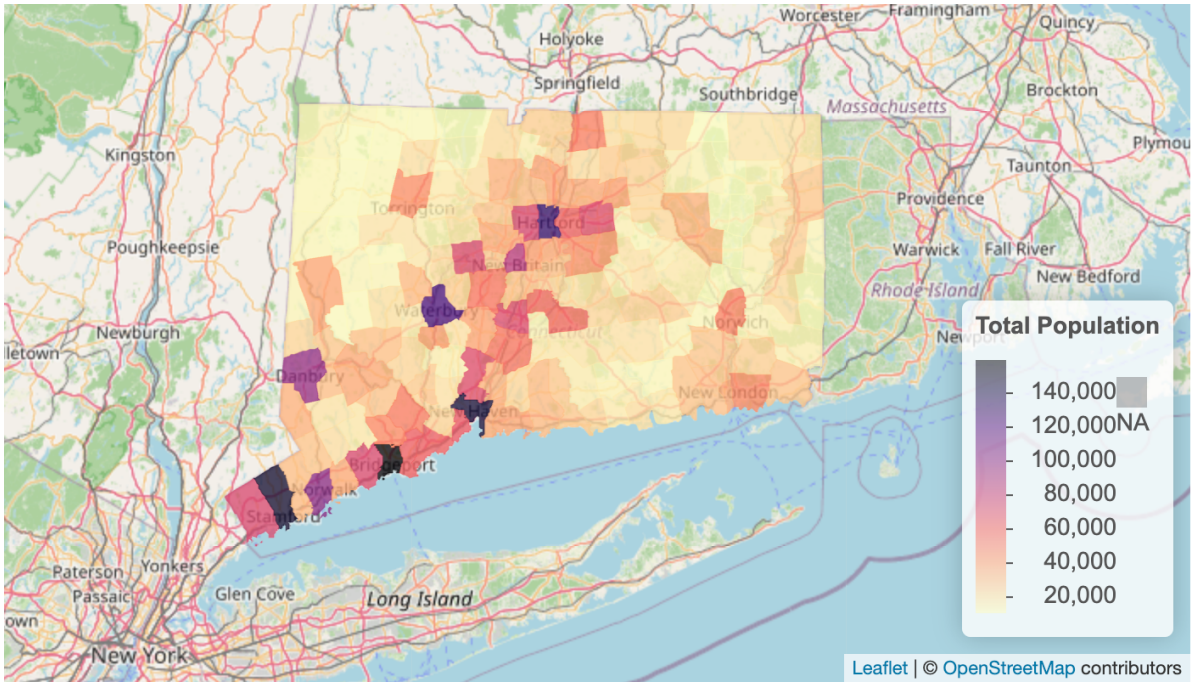


Figure 2: Total Population of Towns in Connecticut in 2024

that makes the Census Bureau's own tools hard to navigate and aligns the data with the geographic units that Connecticut policy uses. The trend of total population of Connecticut in over the last 19 years is represented by Figure 3.

3 Methods

Methods: What did you actually do? What techniques or methods did you employ? What were the specifications for any statistical models you used? What software or packages did you use or develop?

4 Results

Results: What did you learn about the problem you identified in Section 1? This is where you put the tables, figures, and analytics by-products of your work.

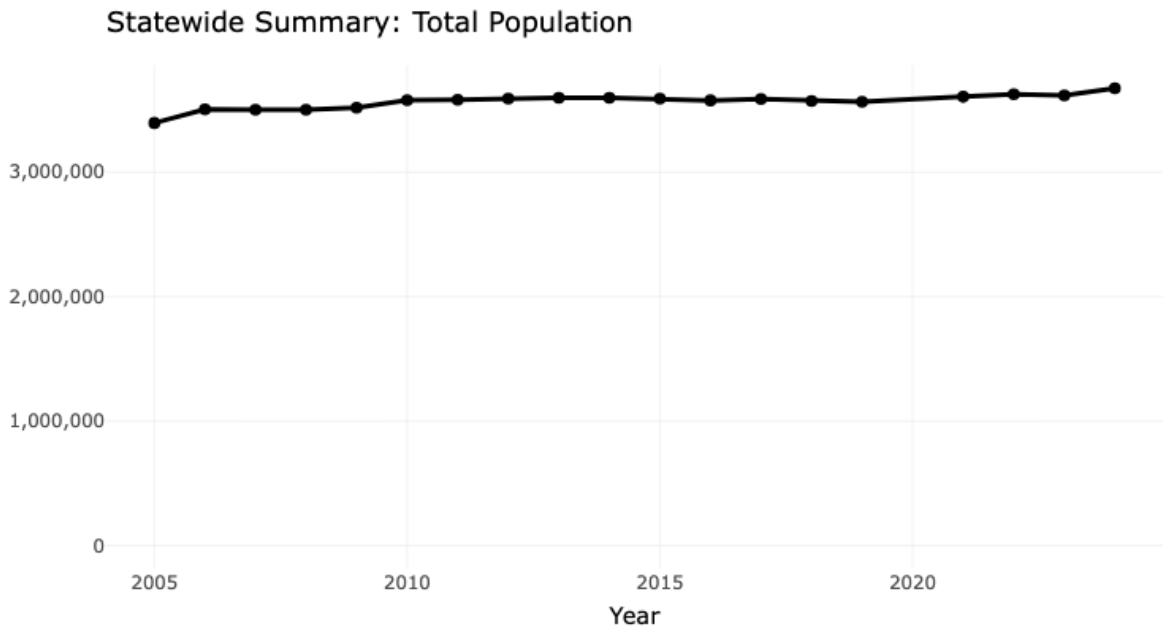


Figure 3: Connecticut Total Population 2005-2024

5 Conclusion

Conclusion: What are the limitations of your work? What are some next steps that someone (either you or another research group) should consider in attempting to further your work? Remind us one last time about what you did.

5.1 Limitations

5.2 Future work

5.3 Final thoughts

6 Ethical statement

Drawing on what you have learned about data science ethics in this class, discuss any ethical considerations in your project. For some projects, this statement could be quite short (one paragraph may suffice). For other projects, more detail may be needed (no more than 2 pages).

Be expansive and creative in your thinking about **possible** ethical considerations. One way to do this assignment poorly would be to write a short statement asserting

that there are no ethical considerations, only to have me think of several fairly obvious ones.

7 Acknowledgements

Were there any people or organizations who helped you that you would like to acknowledge?

References

References: Every reference (except for books) needs a DOI (strongly preferred) or URL. Use Google Scholar to help with BibTeX. Every book needs publisher location (e.g. "City, ST: Publisher"). See SDS 100: Bibliographies

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